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$$\text{Opgave: } z + |z| = 5(5 + i)$$

$$\Rightarrow z = (x + yi) \Rightarrow (x + yi) + \sqrt{x^2 + y^2} = 5(5 + i)$$

$$\Rightarrow \left(x + \sqrt{x^2 + y^2} \right) + yi = 25 + 5i$$

$$\Rightarrow \begin{cases} x + \sqrt{x^2 + y^2} = 25 \\ y = 5 \end{cases}$$

$$\Rightarrow x + \sqrt{x^2 + 25} = 25 \Rightarrow x^2 + 25 = (25 - x)^2$$

$$\Rightarrow x^2 + 25 = 25^2 + x^2 - 2 \cdot 25 \cdot x$$

$$\Rightarrow x^2 - x^2 + 50x = 25^2 - 25$$

$$\Rightarrow 50x = 625 - 25 \Rightarrow x = \frac{600}{50} = \frac{60}{5} = 12$$

$$z = 12 + 5i$$

References